

RoyaleCoach

Software Developer & Computer Science Student

Specializing in backend system architecture, cryptographic applications, terminal interfaces, and resilient software design. Focused on building production-grade software with security-first principles.

```
system_info.sh
```

```
$ whoami
```

```
RoyaleCoach <royalecoach57@gmail.com>
```

```
$ location --current
```

```
Surabaya, Indonesia (ITS Informatics)
```

```
$ stack --core
```

```
[Python, Go, Rust, C#, C/C++, SQLite, Linux, Cryptography]
```

INSTITUTION

ITS Surabaya

FACULTY

FTEIC / Informatics

GITHUB

github.com/RoyaleCoach

01 / About & Professional Summary

I am an undergraduate Informatics Engineering student at Institut Teknologi Sepuluh Nopember (ITS) passionate about constructing software that is clean, highly maintainable, secure, and pragmatically useful.

Rather than building surface-level tutorial projects, my engineering focus centers on designing robust systems that solve realistic computational challenges. My primary interests span terminal application design, cryptographic protocols, AI agent architectures, and low-level Linux systems interaction.

02 / Software Engineering Philosophy

Security-First Architecture

Prioritizing data integrity and modern encryption standards (e.g., Argon2id, AES-256-GCM) over legacy algorithms at every design phase.

Minimalist CLI Efficiency

Crafting developer tools and terminal utilities that leverage rich UI capabilities without sacrificing terminal performance.

Clean Architecture & SOLID

Decoupling business logic from frameworks and database adapters to ensure code maintainability, testability, and clarity.

Scalable Domain Systems

Understanding underlying programming paradigms (OOP, functional, memory-safe abstractions) over memorizing syntax.

Core Values

Simplicity • Maintainability • High Performance • Code Readability • Long-term Scalability

03 / Technical Skills & Architecture

PROGRAMMING LANGUAGES

Python

Go

Rust

C#

JavaScript

C/C++

COMPUTER SCIENCE & ARCHITECTURE

Object-Oriented Programming

Algorithms & Data Structures

Clean Architecture

SOLID Principles

Design Patterns

System Design

SECURITY & CRYPTOGRAPHY

AES-256-GCM

Argon2id Key Derivation

Authenticated Encryption

Secure File Handling

Metadata Headers

ARTIFICIAL INTELLIGENCE SYSTEMS

AI Agent Architecture

Hierarchical Memory

Reflection Systems

Prompt Engineering

OPERATING SYSTEMS & ENVIRONMENTS

Linux (Debian-based)

Shell Scripting

Terminal Workflow

POSIX Standards

DEVELOPER TOOLS & LIBRARIES

Git & GitHub

VS Code

SQLite

Typer CLI

Rich

JSON Storage

04 / Featured Project: Cryptid

PRODUCTION-GRADE CRYPTOGRAPHIC TOOLKIT

Cryptid CLI

Cryptid is a modern command-line cryptographic toolkit engineered with security-first principles. Built to replace obsolete or insecure cryptographic tools, Cryptid provides authenticated encryption and modern password hashing algorithms wrapped in an intuitive, performant terminal interface.

TECHNOLOGIES

Python • Typer • Rich • AES-256-GCM • Argon2id • Cryptography Standards

Key Architectural Features

- **AES-256-GCM Encryption:** Ensures both confidentiality and authenticity, preventing ciphertext tampering.
- **Argon2id Key Derivation:** Utilizes memory-hard hashing to resist GPU and ASIC brute-force attacks.
- **Secure Metadata Headers:** Embeds cryptographic salt, initialization vectors, and flags safely inside file headers.
- **Robust CLI Interface:** Provides progress tracking, interactive password prompts, and detailed error messages via Rich and Typer.

```
cryptid_demo.sh
```

```
$ cryptid encrypt --input secret.pdf --out secret.enc
[+] Deriving key with Argon2id (memory=64MB, parallelism=4)...
[+] Encrypting payload with AES-256-GCM...
[SUCCESS] File encrypted successfully -> secret.enc
```

05 / Featured Projects: RPG-Game & NoteApp

TERMINAL-BASED ARCHITECTURE DRIVEN SYSTEM

RPG-Game Engine

A terminal-based RPG game that prioritizes solid software architecture, object-oriented principles, and state persistence over visual graphics.

- Turn-based combat system with dynamic turn management and shield mitigation mechanics.
- Modular skill tree, inventory, item rarity classifications, and character progression state.
- SQLite database persistence storing game state, inventory items, and character stats cleanly.

Stack: **Python • SQLite • Object-Oriented Architecture**

DESKTOP PRODUCTIVITY APPLICATION

NoteApp

A structured note management desktop application built around Clean Architecture principles to keep user data decoupled and easily accessible.

- Hierarchical category and multi-tag indexing system for instant search filtering.
- Automatic background state persistence preventing data loss during session interruptions.
- Lightweight JSON storage layer structured around clean domain-repository patterns.

Stack: **Python • Clean Architecture • JSON Storage • Desktop GUI**

06 / Terminal Utilities & Portfolio Web

COMMAND-LINE UTILITY

QRGen

A lightweight command-line QR code generator designed for terminal environments with dynamic rendering backends.

- Native UTF-8/Unicode block rendering with automatic ASCII fallback for legacy shells.
- Dynamic terminal geometry detection to prevent output line-wrapping artifacts.
- Support for exporting generated matrix data into image and raw text formats.

Stack: **Python** • **CLI** • **Terminal Geometry API**

CURRENT ACTIVE WEB PROJECT

Personal Portfolio Website

A modern, interactive engineering portfolio web application built with React to showcase software projects, GitHub metrics, and interactive terminal sessions.

- Animated hero header and embedded interactive web-terminal emulator.
- Dynamic GitHub statistics integration and customized project breakdown viewports.
- Fully responsive layout built with modern minimalist UI design principles.

Stack: **React** • **JavaScript** • **CSS3** • **GitHub REST API**

07 / Learning Journey Timeline

● 2024 — Systems Exploration

Expanded horizons beyond Python into low-level and compiled software development paradigms.

- Mastered syntax & memory paradigms of **Go**, **Rust**, **C#**, and **JavaScript**.
- Shifted focus from memorizing syntax to evaluating language execution models and safety guarantees.

● 2025 — Applied Architecture & Linux

Constructed multi-layered software applications and migrated fully to Linux daily-drive environments.

- Built **RPG-Game**, **QRGen**, and **NoteApp**.
- Complete personal workflow migration to **Linux (Debian)** & POSIX shell tooling.
- In-depth study of Clean Architecture, SOLID design principles, and SQLite database modeling.

● 2026 — Cryptography & AI Agent Architecture

Advanced systems engineering, cryptographic security implementation, and higher education.

- Engineered **Cryptid** security toolkit with Argon2id and AES-256-GCM.
- Researching **AI Agent Architectures**, including Hierarchical Memory & Reflection Engine designs.
- Undergoing Informatics Engineering degree at **ITS Surabaya (FTEIC)**.

08 / Future Research & Contact

Future Engineering Exploration

- **AI Agent Reflection Systems:** Implementing self-correcting agent loops with persistent vector memory trees.
- **Zero-Knowledge & Advanced Cryptography:** Extending Cryptid with asymmetric key exchange (ECC) and ZK-proofs.
- **High-Performance Systems in Rust:** Developing zero-cost abstraction network services and kernel-space utilities.

Let's Build Something Clean & Secure

Open for software engineering internships, collaborative open-source projects, and technical research networking.

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GitHub: github.com/RoyaleCoach

Location: Surabaya, East Java, Indonesia

University: Institut Teknologi Sepuluh Nopember (ITS)